

DISCLOSURE STATEMENT

I am not aware of any affiliations, memberships, funding or financial holdings that might be perceived as affecting the objectivity of this presentation

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The background of the slide features a dark, swirling pattern of green and yellow smoke or vapor, resembling cannabis smoke. In the bottom right corner, there are several green cannabis leaves.

The effects of cannabis and other illicit substances

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Plan

- Use of cannabis and other drugs in the UK
- The effects of long-term use on health and functioning
 - Links between cannabis and psychosis
- Signs of addiction
- Treatment and management

Use of cannabis and other drugs in the UK

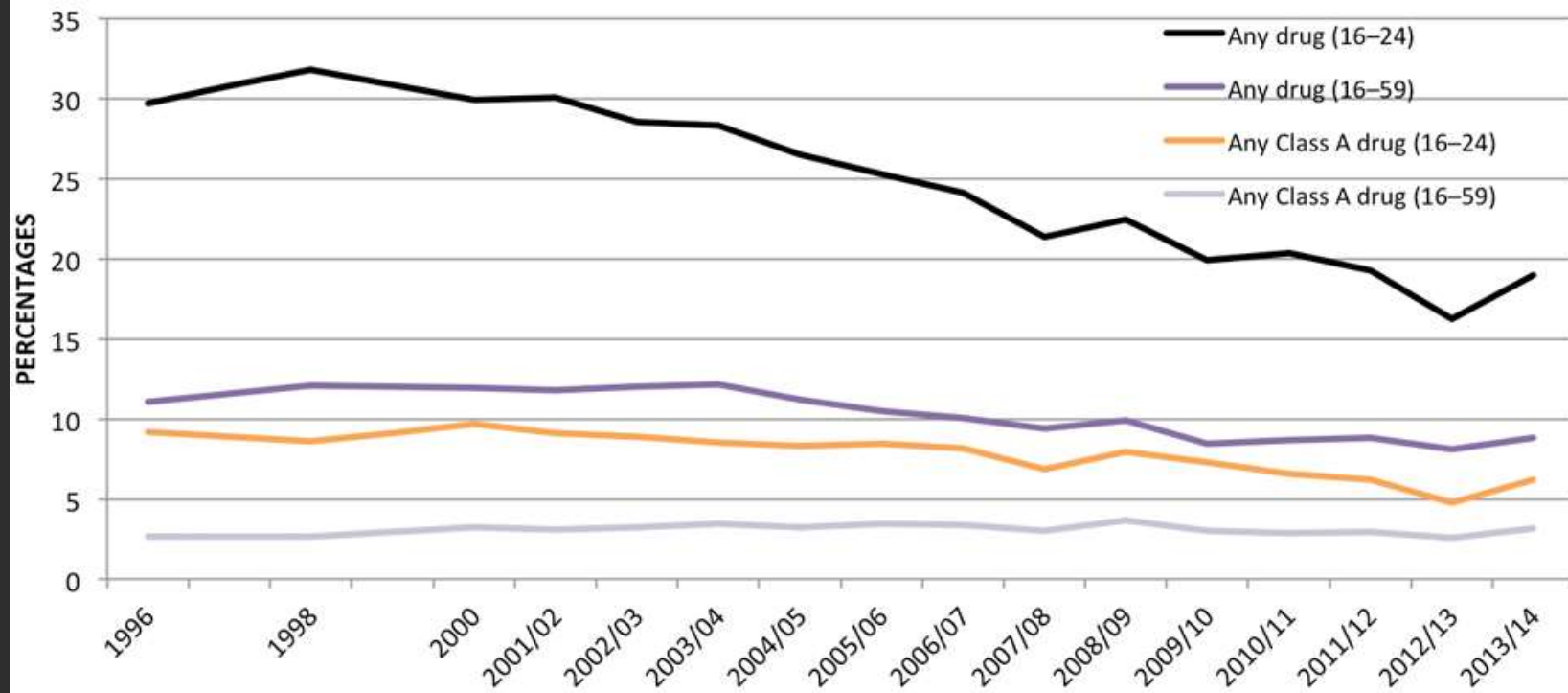


- Nearly 1/3 (31%) of adult population in the UK has taken illegal substances, cannabis being the most used substance with 93% (British Drug Survey, 2014)
- Half of all drug seizures worldwide are cannabis seizures (United Nations report, 2012)

What are the characteristics of drug using groups?

- Young
- Male - Around one-in-eight (11.8%) men had taken an illicit drug in the last year, according to the 2013 to 2014 Crime Survey for England and Wales, compared with 5.8% of women
- People living in urban areas reported higher levels of drug use than those living in rural areas
- Higher levels of drug use are associated with increased frequency of visits to pubs, bars and nightclubs (i.e. visiting nightclub at least four times in the past month)

Illicit drug use in the UK-1996 to 2013 to 2014



2013 to 2014 Crime Survey for England and Wales (CSEW)



Cannabis use

- About three-in-ten adults (16-59 yr olds) reported using cannabis at some point during their lifetime (29.9%)
- Younger adults (16- 24 yr olds): cannabis was again the most used drug (30.8%)

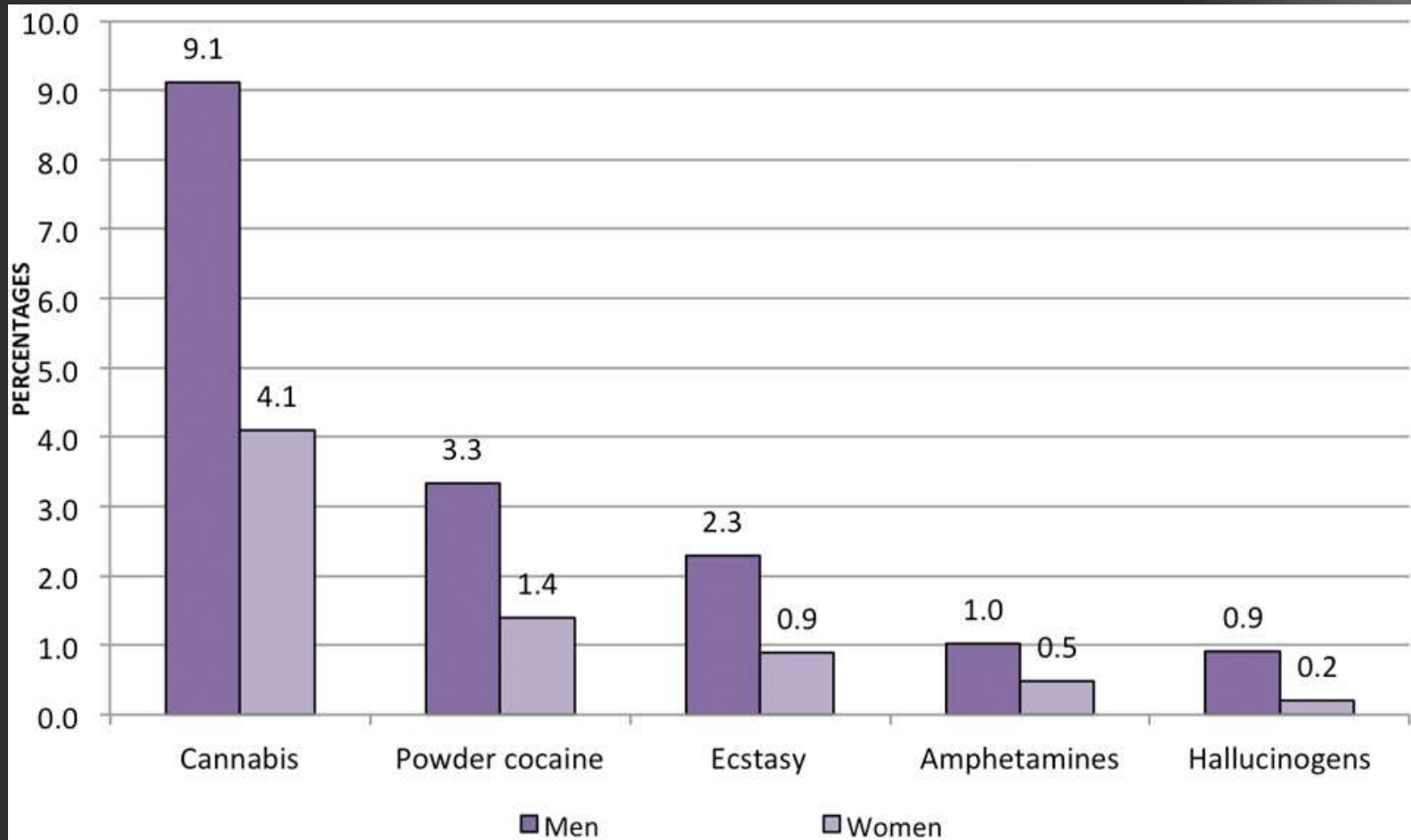
Cocaine use

- After cannabis, cocaine powder was the next most commonly used drug in the last year by adults aged 16 to 59 (2.4%) (British Drug Survey, 2014)
- Although the proportion of adults reporting using powder cocaine in 2013 to 2014 was much higher than in 1996 (0.6%), usage still remains lower than the peak in 2008 to 2009 (3.0%) in 16 – 59 year olds

Ecstasy use

The level of ecstasy use by adults aged 16 to 59 has remained between 1.2% and 2.1% since in 1996

2013 to 2014 Crime Survey for England and Wales (CSEW)

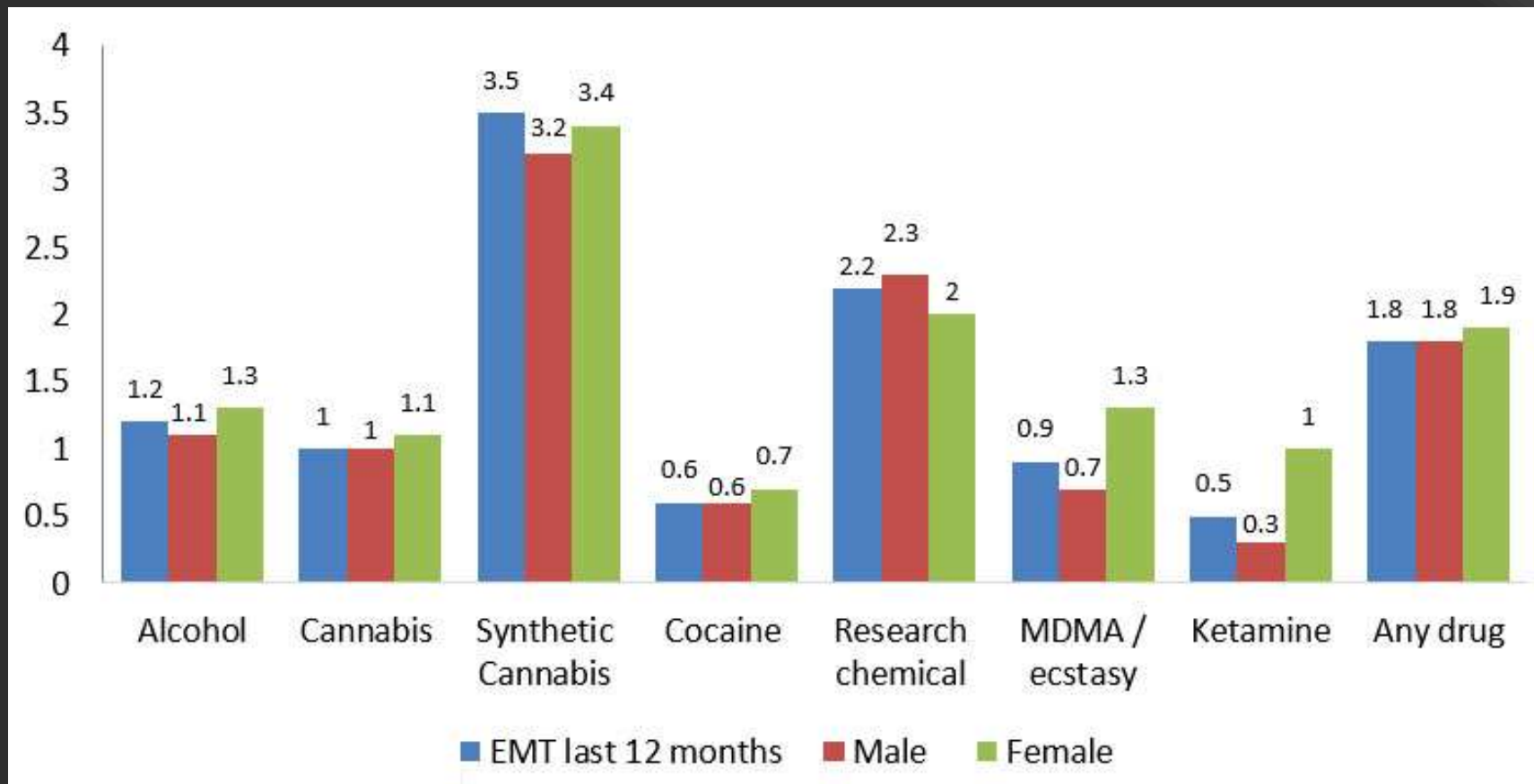


Novel psychoactive substances (NPS) and synthetic cannabis (Spice/Legal Highs)

- UK has the highest rate of NPS purchase and most buy it online
- NPS users are 3 times more likely to seek emergency treatment than the users of other traditional drugs
- Synthetic cannabinoids more likely to lead to emergency treatment than any other drug
- Overall risk of seeking emergency treatment when using Spice is 30 times greater than skunk
- Over half of regular users try to stop use, but report withdrawal symptoms (Global Drug Survey, 2016)



% of people seeking emergency medical treatment following use of drugs and alcohol in the last 12 months – Global Drug Survey, 2015



Changing times..

Possession – legal:

- Uruguay
- The Netherlands
- USA states; Colorado and Washington

Medicinal use – legal:

- Canada
- Czech Republic
- Israel
- Bangladesh
- North Korea
- 14 States in the USA

Uruguay is the first country to legalize the sale, cultivation and distribution of cannabis



So what is so different about cannabis ?



Cannabis plant

A complex plant with over 60 compounds and 400 chemicals. There are four major compounds:

- Delta-9-tetrahydrocannabinol (THC) 🌿
- Cannabidiol (CBD)
- delta-8-tetrahydrocannabinol 🌿
- Cannabinol 🌿

🌿 = Has psychogenic effect



Effect	THC	CBD	References
Antinflammatory	+	+	Juknat et al, 2011
Immunomodulatory	+	+	Costa et al, 2007
Anticonvulsant	+	+ +	Wallace et al, 2001
Muscle relaxant	+ +	+	Lakhan & Rowland, 2009
Anxiolytic	$\pm$	+ +	Zuardi & Guimaeres, 1997, 2012; Crippa et al, 2009
Psychotropic	+ +	-	D'Souza et al, 2004; Atakan et al, 2013
Antipsychotic	-	+ +	Moreira & Guimaeres, 2005; Leweke et al, 2008
Short-term memory problems	+	-	Hayakawa et al, 2008; Morgan et al, 2010
Neuro-protective Antioxidant	+	+ +	Juknat et al, 2011

THC vs. CBD – acute effects in the brain

Our group found;

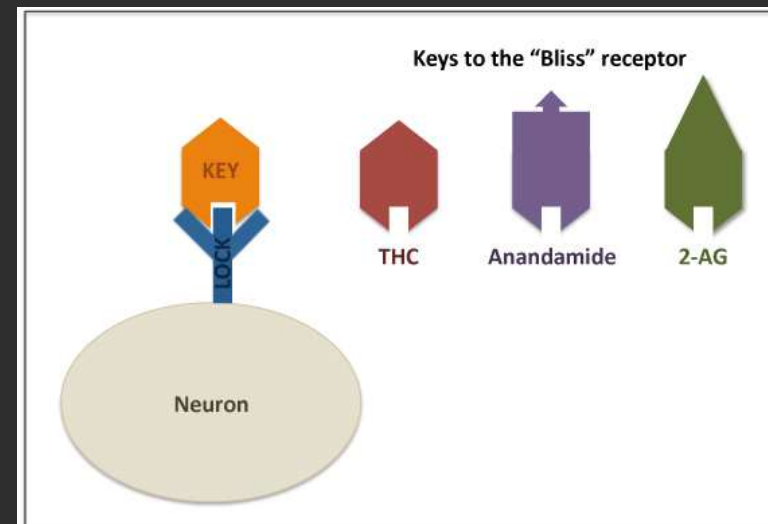
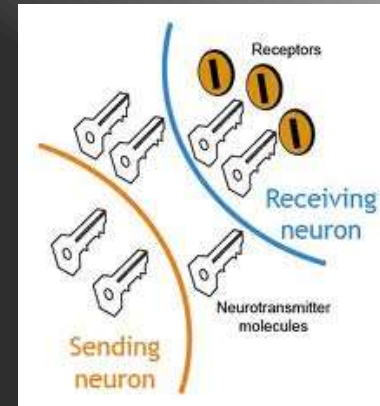
- fMRI studies: **THC and CBD showed opposite effects** on brain activation during cognitive and emotional tasks (Bhattacharyya et al. 2010, 2012; Borgwardt et al. 2008; Fusar-Poli et al. 2009, 2010; Winton-Brown et al. 2011)
- When CBD was given before THC, it prevented the psychotomimetic and anxiogenic effects of THC (Bhattacharyya et al. 2010, Fusar-Poli et al. 2010)
- CBD inhibited acute THC-induced psychotic symptoms as well as THC-induced cognitive impairment (Morrison et al, 2009; Englund et al. 2013)

Endocannabinoid system - history

- In 1988 a specific cannabinoid receptor was found for the first time (CB1 and CB2)
- In 1992: The discovery of the neurotransmitter, a brain molecule called “anandamide” was identified, from a Sanskrit word “ananda” meaning “bliss”

Receptors = Locks
Neurotransmitter = Keys

- Multiple keys can activate the same lock



Endocannabinoid system

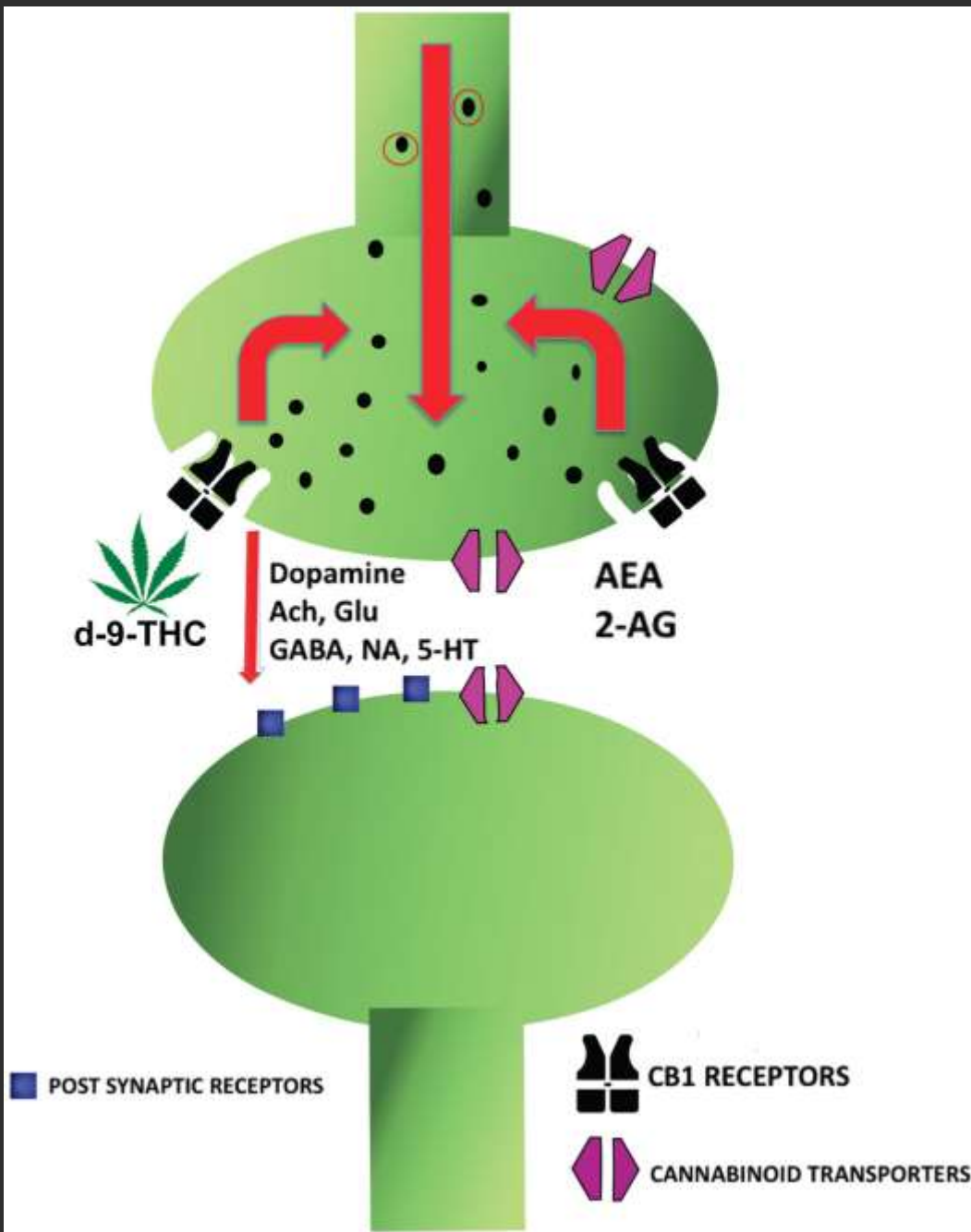


- Found from humans to fruit flies to tuberculosis microbes
- This suggests that the gene must have been present early in evolution, and its conservation implies that the receptor serves important biological functions



- This system is widely present in the brain, spinal cord, liver, heart, thyroid, uterus, bones and genital organs and placenta





When cannabis is used, THC binds to CB1 receptor and acts in a less selective manner

Therapeutic potential of this system

Svizenska et al, 2008

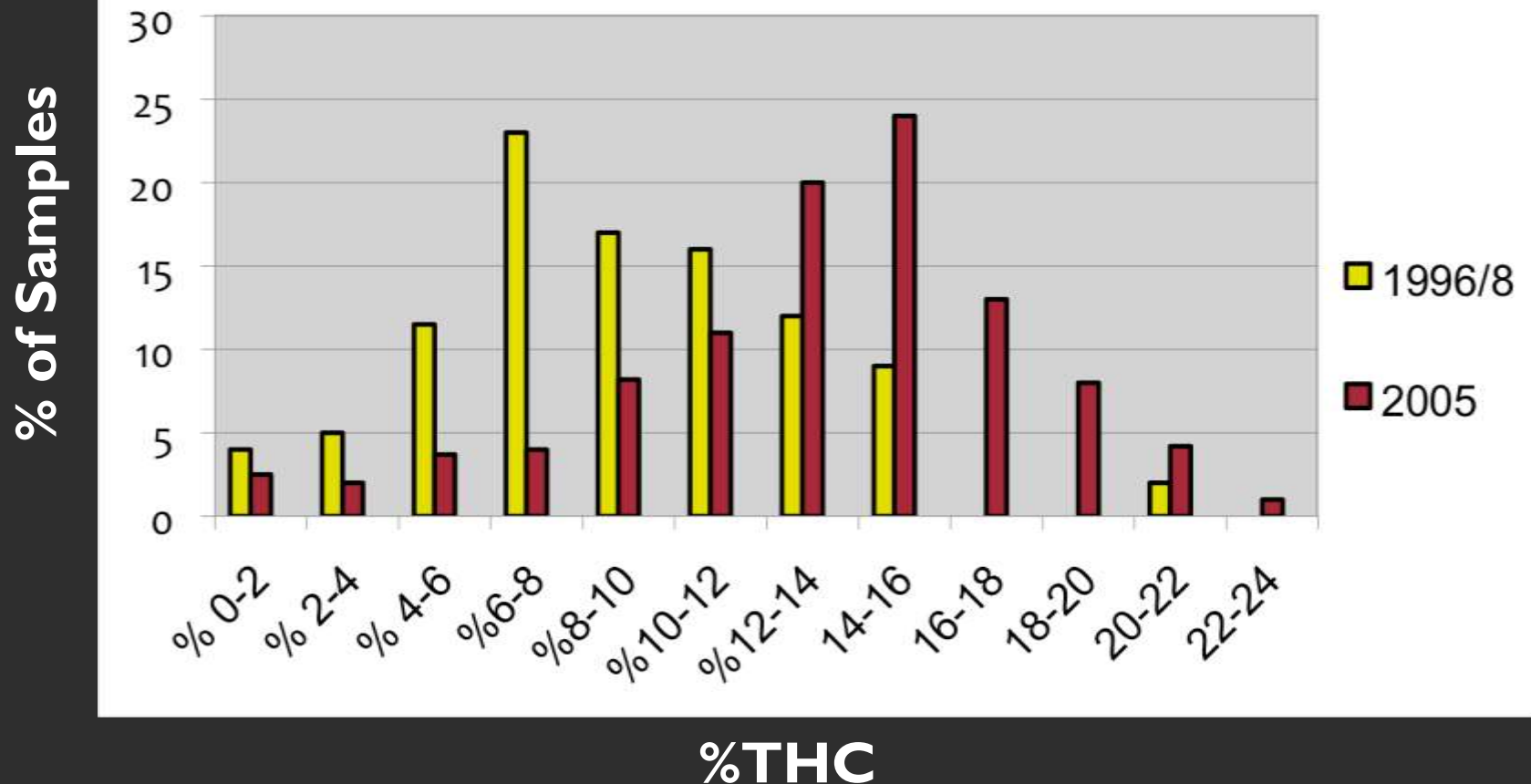
- Hypnotics
- Analgesics
- Anti-emetics
- Anti-asthmatics
- Anti-hypertensives
- Immunomodulatory drugs
- Neuroprotective agents
- Spasticity and other “movement disorders”
- Anti-glaucoma
- Eating disorders
- Alcohol withdrawal
- Bone growth
- Atherosclerosis
- Anti-epileptics
- Anti-psychotics

How important is the strength
of street cannabis?



Sinsemilla potency comparison 1996/8 v 2005

UK:



David Potter et al, 2008

Potency levels significantly higher in 2005 ($p < 0.0001$)

Rising strength of street cannabis

- Skunk has squeezed out other types of cannabis and has minimal CBD
- The amount of THC in skunk has almost tripled over the last 15 years
- Most street cannabis is now home-grown
- Electrical lighting power yields more flowers, hence leading to higher production
- Seeds sold in the Internet with phrases like; “She’ll lock you to the couch like super glue on steroids!”



(Potter, et al 2008 and 2012)

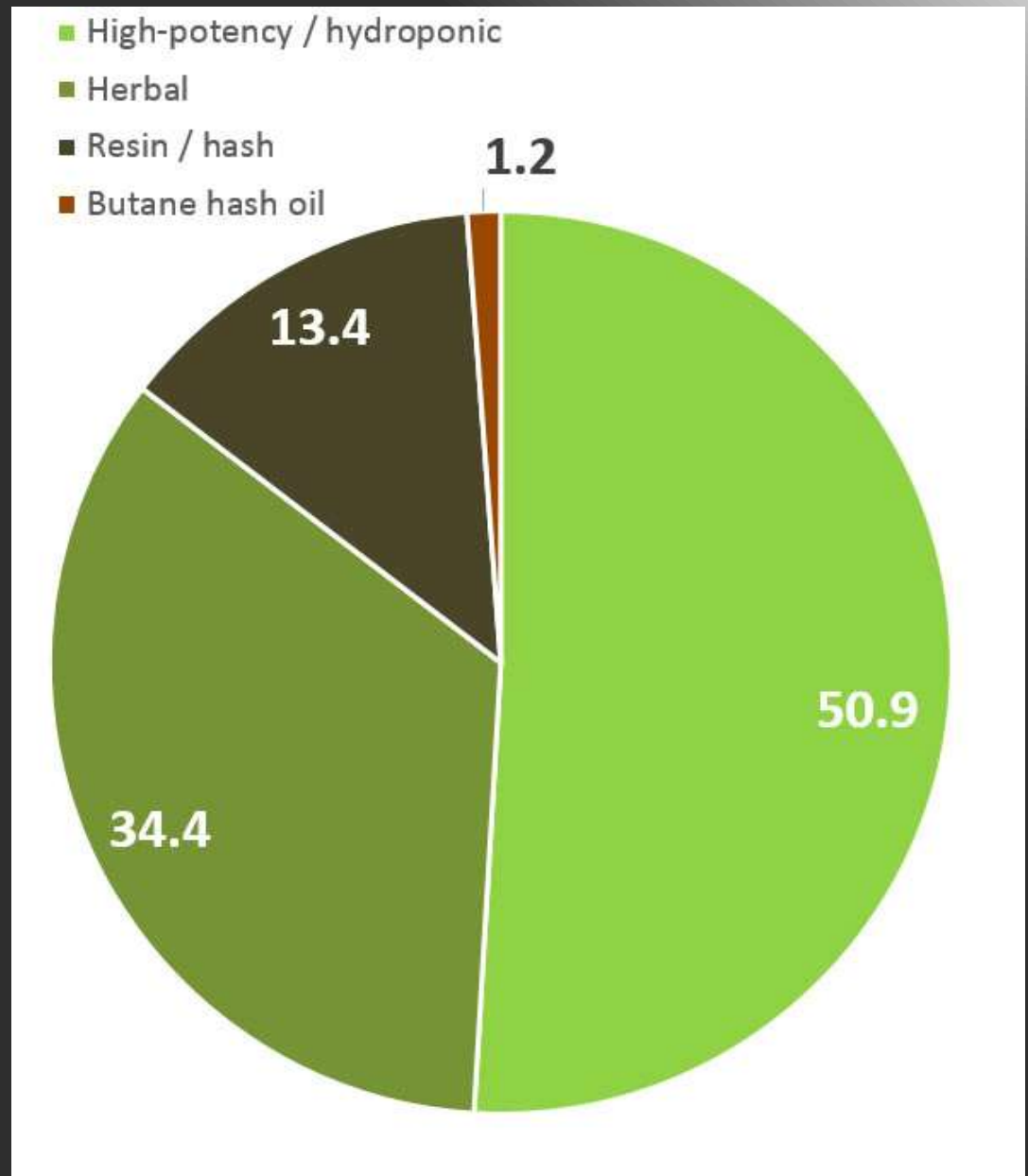
Skunk in South –East London



- In our area (South-East London), using ‘skunk-like’ cannabis has been shown to triple the risk of psychosis
- 24% of new cases of psychosis were attributable to the use of high potency skunk-like cannabis
- The risk increased with the frequency of use and the strength of the variety of cannabis used (Di Forti et al, 2015)

Type of cannabis
used associated
with emergency
medical
treatment during
the last 12
months

1% of people
sought help at
the emergency
clinics, which is a
very high figure
(Global Drug
Survey, 2015)



Cannabis and Psychosis - Existing evidence



- Epidemiological
- Neurobiological
 - Biochemical
 - Neuroimaging
- Genetic studies

Review of longitudinal epidemiological studies; Gage et al, 2016

- Review of epidemiologic studies provides strong enough evidence to warrant a public health message that cannabis use can increase the risk of psychotic disorder
- Further studies are required to examine to identify high-risk groups, particularly susceptible to the effects of cannabis on psychosis

Strong evidence on the following factors for increased risk

Statement	Consistency of evidence	Evidence from E NB G	Evid. grade
Younger age use associated with higher risk	+ + +	E NB	A+
THC interferes with synaptic plasticity	+ + +	NB	A
Those with familial risk have higher sensitivity	+ +	NB G	A
THC/CBD critical	+ +	E NB	A
Early life events and cannabis use	+	E NB	B

E=Epidemiological; NB= Neurobiological; G=Genetic

van Winkel et al, 2014

Other effects of long-term use of cannabis

Links between the following consequences have also been suggested, but the causal relationship has not yet been determined

- Depression
- Anxiety
- Other substance use disorders
- Increased risk of bronchitis and chronic cough

Other effects of long-term use

With cumulative life-time use

- Lower scores on a test of verbal memory
- Negative effects on attention, memory and learning can last for days or weeks after the acute effects of the drug wear off

Conclusions..



- Using cannabis is neither necessary nor sufficient to cause a psychotic illness
- However, exposure to cannabis is most likely to be a component factor which interacts with other factors for the development of an enduring psychotic illness, in those who are genetically at risk
- Links between cannabis use and other mental health problems require further research evidence

What are signs of addiction?

Three Cs of addiction

- Loss of **control** over the amount of use
- **Craving** and compulsive using
- **Continued use** in the face of adverse consequences

Developing tolerance

- Needing to use more amounts in order to get the desired 'stoned' effect or 'high'

What are signs of addiction?



Physical signs

- Being over or underactive (according to the drug used)
- Dilated pupils
- Weight loss
- Excessive sniffing not due to cold
- Looking pale and undernourished

What are signs of addiction?

Behavioural signs

- Greater absenteeism
- Missing important engagements
- Problems at work
- Increased risk of accidents or injuries at work
- Being secretive about activities
- Having financial problems



What can be done?

What can be done?

Assessment

- Why?
- How?
- How much?
- How often?
- What strength?
- What does it do?

Psychoeducation on substance use:

- Campaign and education on the health effects of substances at the primary and secondary school level, similar to tobacco smoking
- Provision of information for the public (leaflets, websites, health services, voluntary organisations and media)
- Provision of treatment programmes for cannabis and other drug use; harm reduction or cessation – extremely limited
- More substance specific information, rather than general

Psychological interventions:



- There is some evidence, based on a few randomized controlled trials on out-patient groups that cognitive behaviour therapy, motivational intervention and a hybrid of other combined therapies can be superior to routine care

(Barrowclough *et al*, 2001; Haddock *et al*, 2003; James *et al*, 2004)



Not thinking about it
(Pre-contemplation)



Thinking about it
(Contemplation)



Planning it
(Preparation)



CYCLE OF CHANGE



Keep doing it
(Maintenance)



Can't do it
(Relapse)



Doing it
(Action)

How to help to those who use cannabis or other drugs?

SMART goal-setting principles to make a change:

- Specific
- Measurable
- Achievable
- Realistic
- Time frame



“I need to reduce my smoking” vs.

“I will smoke one, not two joints tomorrow morning, before breakfast”

Therapeutic skills in MI

OARS is an acronym for the basic therapeutic skills which are used in MI

- **Open questions:** Often start with 'why', 'what' or 'how' (to avoid 'yes' or 'no' replies)
- **Affirmations:**
- **Reflective listening**
- **Summarising**



Therapeutic skills in MI

- Open questions:

Do you like smoking? vs. How do you feel about your smoking?

- Affirmations: Praising client's thoughts for positive change
- Reflective listening: Listening carefully and selectively repeating them back the important parts:
So on the one hand you are worried about how much you smoke cannabis and about how suspicious you become right after and at the same time smoking is one of your few pleasures in life

Therapeutic skills in MI



- **Develop discrepancy:** to draw attention to the difference between where they *are now* and where they ideally want to be
- **Avoid argument:** The harder you argue and try to force someone to change, the more they are to increase their sustain talk
- **Roll with resistance:** Is recognising resistance but not opposing it:

Client: *What do you think I should do about my smoking cannabis? It is the most enjoyable thing in my day*

Therapist: *I think you know more about your situation than me. You are pretty good at coming up with ideas about what to do. I just wondered what ideas you have?*

Maintaining cessation

- Improving social skills
- Replacement of meaningful activity with drug use rituals

Conclusions..

- Early signs of addiction vary according to the drug used
- Absenteeism, problems at work, financial problems can be early signs of addiction
- Even though there are some intervention models to manage drug use, these are still at the developmental stage and services are extremely limited particularly for cannabis use





Thank you!